

Benchmarking Univariate Fay–Herriot Estimates

Region-level ratio adjustment and bootstrap MSE implemented in `bench_regional()` (UFH)

Ifeanyi Edochie and Nobuo Yoshida, World Bank Group

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1. Purpose

This note documents the benchmarking option for the univariate Fay–Herriot (UFH) model in the EU SAE package. It explains why model-based small-area estimates are benchmarked, sets out the benchmarking method, and describes how the mean squared error (MSE) of the benchmarked estimates is computed. The exposition is grounded in two sources: the statistical methodology of the `emdi` R package, which the dashboard uses to fit the UFH model and which itself follows Datta, Ghosh, Steorts, and Maples (2011); and the package’s own implementation in `scripts/ufh_functions.R`, function `bench_regional()`, invoked from `scripts/01_ufh.R`. The UFH benchmarking routine is the single-survey, single-period analogue of the multivariate routine `bench_regional_mfh()` documented in the companion MFH note; readers familiar with that note will recognize the same ratio adjustment and the same bootstrap logic, here without the cross-time terms.

2. Notation

Let $d = 1, \dots, D$ index the small areas (domains), of which $m \leq D$ are observed in the survey (in-sample) and $D - m$ are out-of-sample. Write $\hat{\theta}_d^{\text{dir}}$ for the direct survey estimate of the poverty rate in domain d , ψ_d for its (known) sampling variance, and x_d for the vector of auxiliary covariates. Let θ_d denote the unobserved true area value, $\hat{\theta}_d^{\text{FH}}$ the Fay–Herriot EBLUP — the same quantity the companion MFH note writes $\hat{\theta}_{dt}^{\text{EBLUP}}$, stored in the FH column of the output — and $\hat{\theta}_d^{\text{Bench}}$ its benchmarked counterpart. Benchmark areas (the administrative units that supply the constraints — regions by default) are indexed r , and N_d is the population size of domain d . The notation follows the MFH note (domains $d = 1, \dots, D$, benchmark areas r) specialized to a single time period, so that note’s time index t is dropped here.

3. The univariate Fay–Herriot model and the EBLUP

The package fits the area-level Fay–Herriot model (Fay and Herriot, 1979) with `emdi::fh()`. The model has a sampling layer and a linking layer,

$$\hat{\theta}_d^{\text{dir}} = \theta_d + e_d, \theta_d = x_d^\top \beta + u_d,$$

with independent sampling errors $e_d \sim N(0, \psi_d)$ and area random effects $u_d \sim N(0, \sigma_u^2)$. The default fit uses REML for σ_u^2 . The resulting empirical best linear unbiased predictor is the shrinkage (composite) estimator

$$\hat{\theta}_d^{\text{FH}} = \hat{\gamma}_d \hat{\theta}_d^{\text{dir}} + (1 - \hat{\gamma}_d) x_d^\top \hat{\beta}, \hat{\gamma}_d = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + \psi_d},$$

so that unreliable direct estimates (large ψ_d , small $\hat{\gamma}_d$) are pulled toward the regression synthetic part $x_d^\top \hat{\beta}$, and out-of-sample domains take the synthetic value $\hat{\theta}_d^{\text{FH}} = x_d^\top \hat{\beta}$. The package supports the model on three scales through the `transformation` argument: none, log (welfare means, back-transformed by smearing), and arcsin (rates, back-transformed either by a bias-corrected integral or naively). The unbenchmarked EBLUP's MSE is the analytical Prasad–Rao / Datta–Lahiri estimator under the untransformed and log scales (`mse_type = "analytical"`) and a parametric bootstrap under the arcsin scale (`mse_type = "boot"`).

4. Why benchmark?

The shrinkage that makes the EBLUP efficient also breaks aggregate consistency. Because each $\hat{\theta}_d^{\text{FH}}$ is pulled toward its own synthetic mean, the population-weighted average of the model estimates over a region will in general differ from the corresponding direct regional aggregate — even though the direct aggregate, pooled over many sampled units, is itself design-unbiased and reliable at that coarser level. National statistical offices typically require that disaggregated estimates reproduce the trusted higher-level totals, whether those come from the same survey or from an external official source. Benchmarking enforces this consistency: it rescales the EBLUPs so that they reproduce the regional benchmark exactly, while preserving the relative pattern of poverty across domains within each region.

5. The benchmarking framework

Benchmarking solves a constrained adjustment problem. Given EBLUPs $\hat{\theta}_d^{\text{FH}}$, non-negative weights ω_d that sum to one over the benchmarked set, and a target τ , the benchmarked estimates $\tilde{\theta}_d$ minimize a weighted distance from the EBLUPs subject to reproducing the target,

$$\min_{\{\tilde{\theta}_d\}} \sum_d \phi_d (\tilde{\theta}_d - \hat{\theta}_d^{\text{FH}})^2 \text{ subject to } \sum_d \omega_d \tilde{\theta}_d = \tau.$$

The Lagrange solution is $\tilde{\theta}_d = \hat{\theta}_d^{\text{FH}} + (\omega_d / \phi_d) (\tau - \sum_j \omega_j \hat{\theta}_j^{\text{FH}}) / \sum_j (\omega_j^2 / \phi_j)$. The `emdi` package's `benchmark()` function (Kreutzmann et al., 2019), following Datta et al. (2011), exposes two choices of the internal weights ϕ_d , which give the two standard estimators in the table below.

Table 1. Benchmarking types in `emdi::benchmark()`

TYPE	INTERNAL WEIGHT ϕ_d	RESULTING BENCHMARKED ESTIMATE
Raking	ω_d	$\tilde{\theta}_d = \hat{\theta}_d^{\text{FH}} + (\tau - \sum_j \omega_j \hat{\theta}_j^{\text{FH}})$

Ratio	$\omega_d / \hat{\theta}_d^{\text{FH}}$	$\tilde{\theta}_d = \hat{\theta}_d^{\text{FH}} \tau / \sum_j \omega_j \hat{\theta}_j^{\text{FH}}$
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Raking applies the same additive shift to every domain; ratio applies a common multiplicative factor. For poverty rates — non-negative proportions — the package uses the ratio form, because a multiplicative adjustment cannot drive any domain below zero and it preserves the proportional ordering of the EBLUPs. The package generalizes the single-constraint `benchmark()` to multiple benchmark areas by applying the ratio adjustment independently within each area, exactly as the MFH routine does.

6. Point-estimate benchmarking in the package

The function `apply_regional_bench()` inside `bench_regional()` implements the ratio adjustment one benchmark area at a time. Within each area r it forms the target from the **direct** estimates and the model-side mean from the EBLUPs, both population-weighted over the same domains:

$$B_r = \frac{\sum_{d \in r} N_d \hat{\theta}_d^{\text{dir}}}{\sum_{d \in r} N_d}, \hat{\theta}_r^{\text{FH}} = \sum_{d \in r} \omega_{r,d} \hat{\theta}_d^{\text{FH}}, \omega_{r,d} = \frac{N_d}{\sum_{j \in r} N_j}.$$

The target B_r is the survey-direct regional poverty rate — design-unbiased at the regional level — so this is an internal benchmark to the direct estimator (an external official total could be substituted). The region’s ratio factor and the benchmarked estimate are

$$\lambda_r = \frac{B_r}{\hat{\theta}_r^{\text{FH}}}, \hat{\theta}_d^{\text{Bench}} = \lambda_r \hat{\theta}_d^{\text{FH}} (d \in r),$$

with λ_r set to 1 whenever B_r is missing or the denominator falls below `eps_denom` = 10^{-8} . By construction the population-weighted mean of $\hat{\theta}_d^{\text{Bench}}$ within region r equals B_r , so the benchmarked map reproduces every regional direct rate exactly. When the data carry no benchmark-area column, all domains are assigned to a single synthetic “National” area and B_r becomes the national population-weighted direct rate.

7. MSE of the benchmarked rates

Following Datta et al. (2011), the MSE of a benchmarked predictor is defined relative to the true small-area value, not relative to the benchmark it was forced to meet:

$$MSE(\hat{\theta}_d^{\text{Bench}}) = E[(\hat{\theta}_d^{\text{Bench}} - \theta_d)^2].$$

Because $\hat{\theta}_d^{\text{Bench}}$ is a non-linear function of the EBLUPs through λ_r , no closed form exists, and the routine `boot_bench_regional()` uses a bootstrap with $B=200$ replicates by default.

The procedure first harvests from the fitted object the variance $\hat{\sigma}_u^2$ (refvar), the coefficients $\hat{\beta}$, the predicted random effects $\hat{u}_d$, the model residuals, and the sampling variances ψ_d . To generate replicate b it draws random effects and residuals nonparametrically from the fitted quantities after centering and rescaling them — the random effects to variance $\hat{\sigma}_u^2$ and the residuals to unit variance — and then forms sampling errors $e_d^{(b)} = \text{res}_d^{(b)} \sqrt{\psi_d}$. This is a *nonparametric* (residual-resampling) bootstrap: the random effects and residuals are drawn with replacement from the rescaled empirical values rather than simulated from a fitted Normal law. It differs in this one respect from the MFH routine, which draws *parametrically* from the fitted $N(0, \hat{\sigma}_u^2)$ law (with the AR(1) recursion for the autoregressive variants); the surrounding logic — re-benchmarking inside each replicate and scoring against the synthetic truth — is identical in the two routines. The synthetic truth and observation, on the untransformed scale, are

$$\theta_d^{(b)} = x_d^\top \hat{\beta} + u_d^{(b)}, y_d^{(b)} = \theta_d^{(b)} + e_d^{(b)} (d \leq m).$$

Each replicate refits the Fay–Herriot model on the synthetic data — by re-evaluating the stored `fh()` call with `update()` on the untransformed scale, or by a manual GLS-plus-EBLUP step on the arcsin scale (which avoids re-applying the transformation to data that are already transformed, with a bias-corrected or naive back-transform). The bootstrap EBLUPs are then **re-benchmarked** by the same ratio adjustment, with $y_d^{(b)}$ playing the role of the direct estimate, producing $\lambda_r^{(b)}$ and $\hat{\theta}_d^{\text{Bench},(b)}$. Re-running the benchmarking inside every replicate is what propagates the sampling variability of λ_r into the MSE. Replicates whose refit fails to converge are skipped and excluded from the count, so the estimate averages the squared deviation from the synthetic truth over the n_{valid} successful replicates — not over the nominal B — matching the MFH routine:

$$\widehat{MSE}(\hat{\theta}_d^{\text{Bench}}) = \frac{1}{n_{\text{valid}}} \sum_{b=1}^{n_{\text{valid}}} (\hat{\theta}_d^{\text{Bench},(b)} - \theta_d^{(b)})^2.$$

When `bench_regional()` is called with `MSE = FALSE`, the benchmarked MSE is set equal to the unbenchmarked FH MSE as a placeholder. After the bootstrap, the FH, benchmarked-FH, and direct estimates are returned to the original scale together — by Duan-style smearing for the log model or the arcsin back-transform for rates — with variances propagated by the delta method, so that `FH_Bench` and its MSE are reported on the same poverty scale as the published map.

8. Function interface and outputs

`bench_regional()` takes the fitted `emdi` model and a small data frame describing the benchmark geography, and returns the model object augmented with the benchmarked column and its MSE.

Table 2. Arguments of `bench_regional()`

ARGUMENT	TYPE	DEFAULT	MEANING
<code>model</code>	<code>emdi fh fit</code>	—	Fitted univariate Fay–Herriot object whose EBLUPs are benchmarked

region_df	data frame	—	Columns domain, region, Nd, direct_povrate defining each area
MSE	logical	FALSE	If TRUE, run the bootstrap to estimate the benchmarked MSE
B	integer	200	Number of bootstrap replicates
seed	integer	123	Seed for reproducibility of the bootstrap draws

The driver `01_ufh.R` builds `region_df` from the survey (the benchmark-area column is read from the variable map and defaults to the region; absent that, a single national area is used), computes the regional benchmarks, calls `bench_regional(..., MSE = TRUE, B = 200)`, and extracts the results with `estimators(fh_bench, MSE = TRUE, CV = TRUE)`. The output table carries `Direct`, `FH`, and `FH_Bench` columns together with their MSEs and coefficients of variation. These feed the dashboard's comparison plot (direct versus FH versus FH-benchmarked), the choropleth of the benchmarked rate `FH_Bench`, and the significance tables produced in both benchmarked and unbenchmarkd form.

9. Practical notes

Benchmarking only redistributes; it does not create information. Its value rests on the benchmark target being more reliable than the individual model estimates, which holds when the regional direct aggregate pools enough sampled units to be precise. Where a region contains only a few small domains, the gain is limited and the analyst should consider a coarser benchmark geography or an external official total. The ratio form was chosen over raking because poverty rates are bounded below by zero; for an unbounded indicator (for example, a log-welfare mean before back-transformation) raking's additive shift may be preferable. Finally, this UFH routine is the single-period analogue of `bench_regional_mfh()`: it shares the ratio adjustment and the “bootstrap against the true value” MSE, but because the univariate model carries no time dimension there is no AR(1) structure and no mean cross-product error (MCPE) — the cross-time quantity documented in the MFH note. When comparable-over-time estimates are required, the multivariate routine should be used instead.

10. References

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